

Question	Answer	Marks
8(a)(i)	circle around both diodes	B1
8(a)(ii)	indicates (whether) temperature	M1
	(is) above or below a set value	A1
8(b)(i)	(when resistance of C > R_V ,) $V^- > V^+$ or $V^+ < 3V$ or p.d. across $R_V < \text{p.d. across } R/Y/3V$ or p.d. across C > p.d. across $R/X/3V$	M1
	op-amp output is negative	M1
	(only) green	A1
8(b)(ii)	resistance of C becomes less than R_V or $V^- < V^+$	B1
	green (LED) goes out	A1
	blue (LED) comes on	A1
8(c)	changes/determines <u>temperature</u> at which LEDs switch	B1

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9(c)(i)	Hall voltage depends on thickness of slice	01